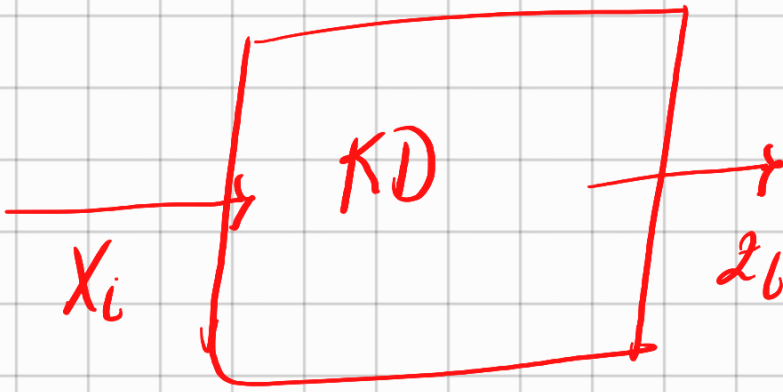


Kombine2on2ol Devreler

Kombine2on2ol t2mle2ik devreler



Medium
scale
integration
(orta 2l2ekli
entegre devre-
leri)

Bunlar
hazır
devrelerdir

1- Aritmetik toplama ve 2ıkarma
devreleri

ve tam toplayıcılar

- yarı ve tam çıkartma
- yarı ve tam çıkartmalar
- toplama ve çıkarma devreleri

2- Multiplexer (Seçiciler)
Mux

3- Demultiplexer (Dağıtıcılar)
DEM

4- Kod Gözücüler (Decoderler)

5- Kodlayıcılar (Encoderler)

6- ALU (Arithmetic logic unit)

7- 7 segment display

1- Aritmetik toplama / çıkarma devreleri

1A) yarı toplayıcı (half adder)

tek digitlik toplama yapabiliyor

		toplama elde	
a	b	T	E
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

a	b	c	d
e	f	g	h

$c + g + e$

elde almayan digit toplamı

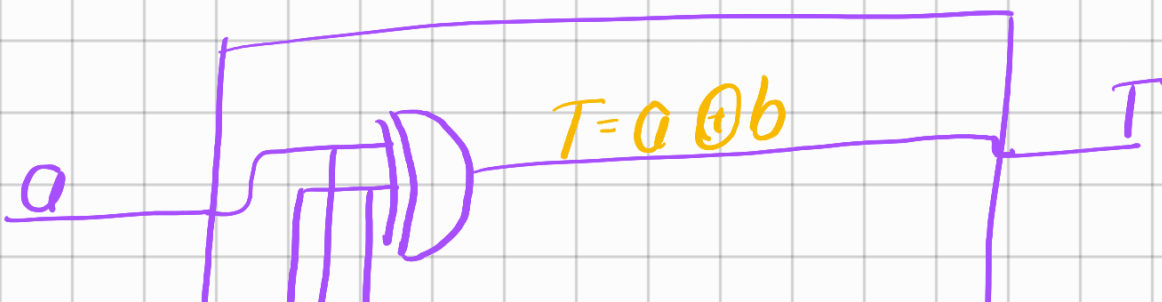
$E = a \cdot b$

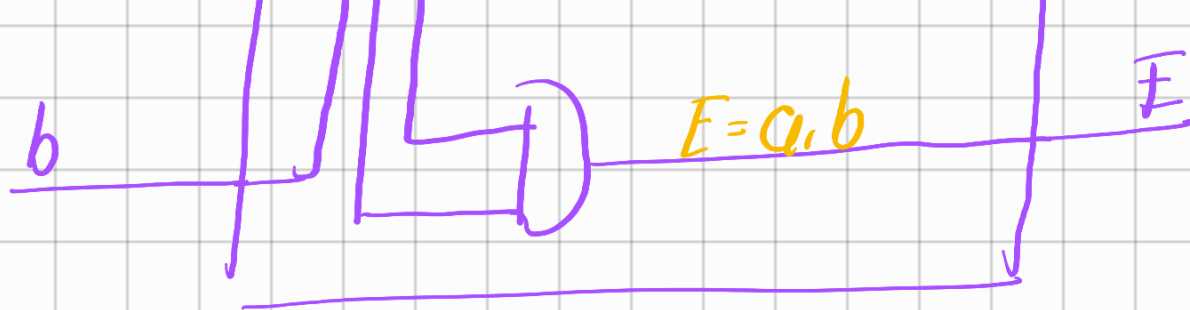
a \ b	0	1
0		1
1	1	

XOR

$T = a'b + b'a$

$T = a \oplus b$





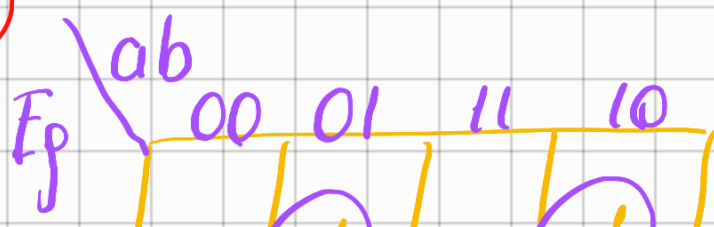
1-B) Tam toplayıcı (full adder) → hem elde-yi alır hem de toplama-yı yapar

E	a	b	toplama elde	
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

(T)

(E)

(T)



0		1		1
1	1		1	

$$1 \rightarrow T = E_g \cdot a' \cdot b' + \bar{E}_g \cdot a' \cdot b' + \bar{E}_g a b + E_g' a b'$$

$$E \rightarrow a \oplus b \oplus E_g$$

	bc	00	01	11	10
a					
0			1		
1		1	1	1	

yani full adder
yapıcağımız
şimdi yaptığımız
2 fonksiyona
bakacağız

$$E = a \cdot b + E_g (a + b)$$

$$C = E_g$$

2 Kapılı XOR

$$\rightarrow a' b + b' a$$

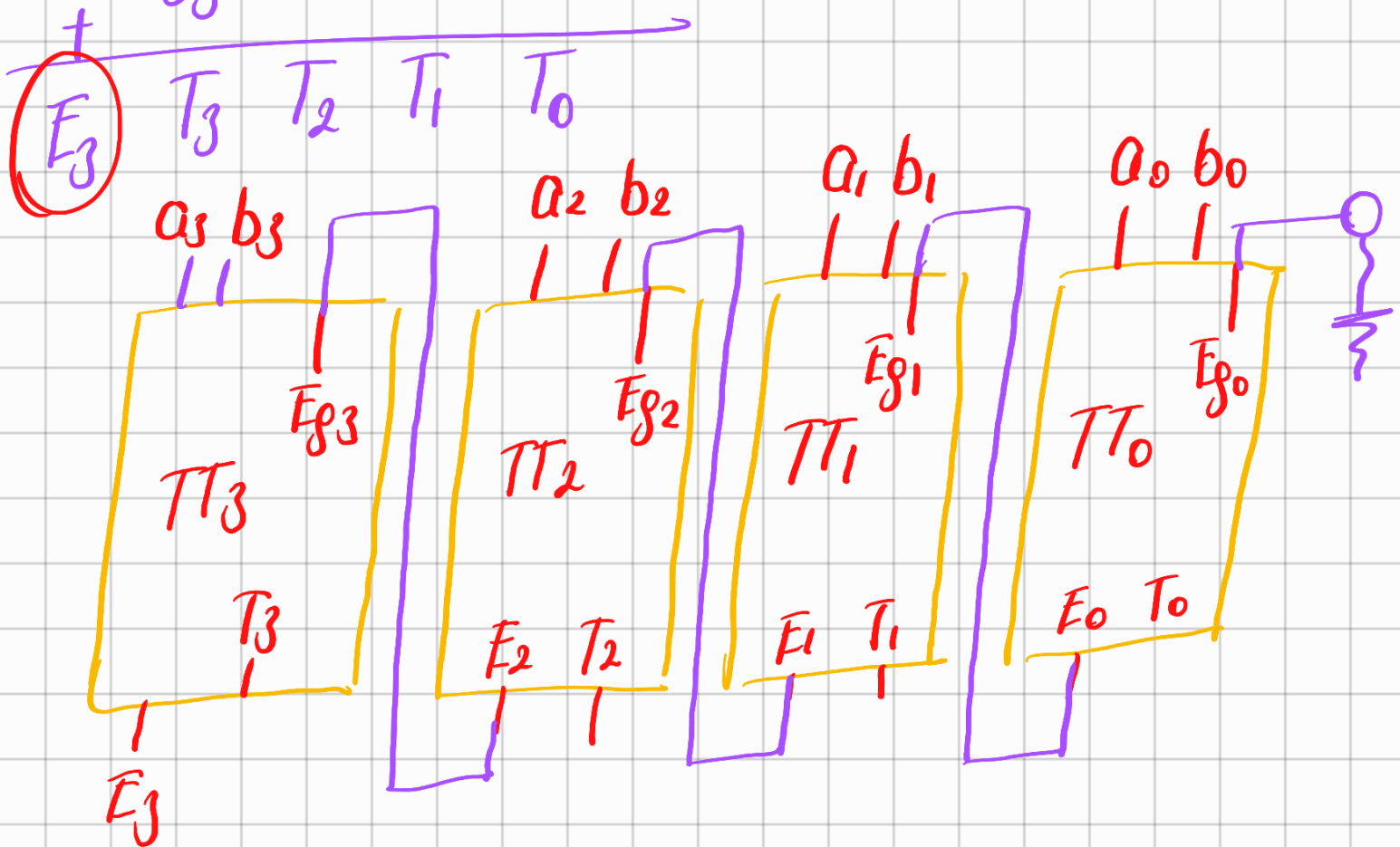
3 Kapılı XOR

$$\hookrightarrow a'b'c + a'bc' + abc + ab'c'$$

Burada bir şey var

ÖR → 4 bitlik 2 sayının toplamını
bulan bir paralel toplayıcı
devresinin tasarımı (TT)

$a_3 \ a_2 \ a_1 \ a_0$
 $b_3 \ b_2 \ b_1 \ b_0$



4 bitlik

paralel
toplayıcı

2A) Yarı Çıkarma (half subtracter)

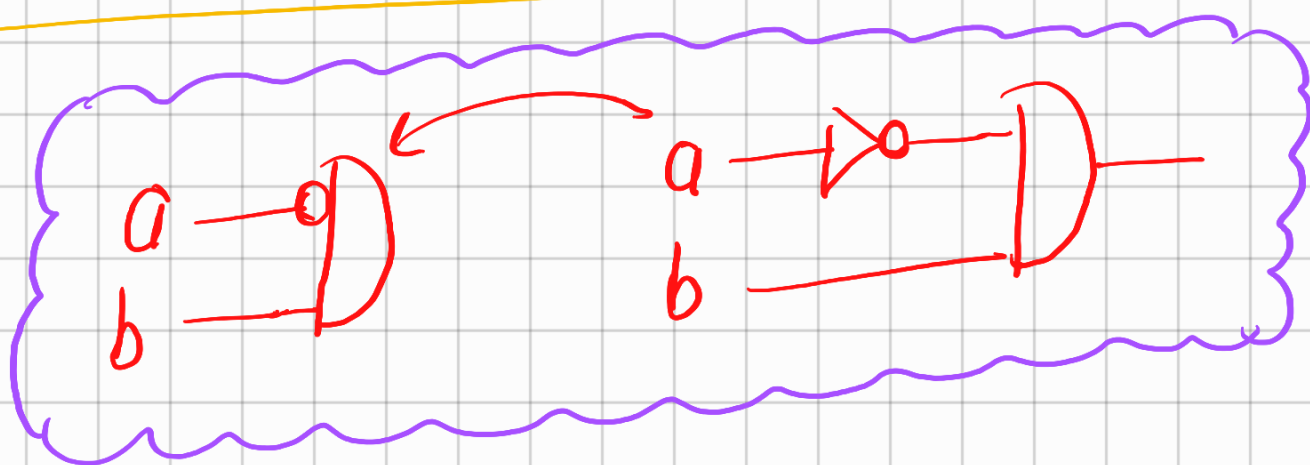
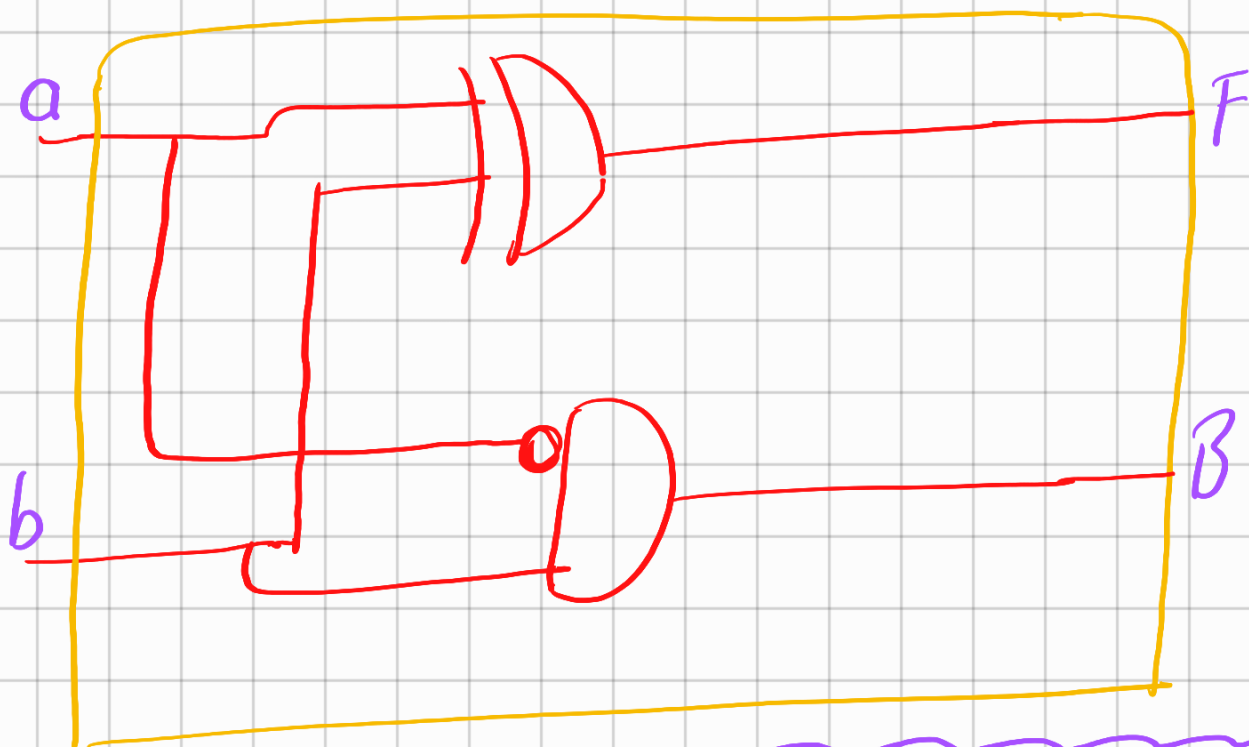
a	b	(fark)	
		F	Borc
0	0	0	0
0	1	1	1
1	0	1	0
1	1	0	0
		(F)	(B)

$$\begin{array}{r}
 3 \\
 \overline{) \begin{array}{r} ab \\ cd \end{array}} \\
 \underline{F} \\
 1 ^2 \\
 \overline{) \begin{array}{r} 10 \\ 01 \end{array}} \\
 \underline{1}
 \end{array}
 \rightarrow \begin{array}{r} 02 \\ 1 \\ \hline 1 \end{array}$$

a \ b	0	1
0		(1)
1	(1)	

$B = a'b$

$F = a \oplus b$



2B) Tam Gıkarıcı (full subtracter)

a	b	B_g	fark F	Borç B
0	0	0	0	0
0	0	1	1	1
0	1	0	1	1

0	1	1	0	1
1	0	0	1	0
1	0	1	0	0
1	1	0	0	0
1	1	1	1	1

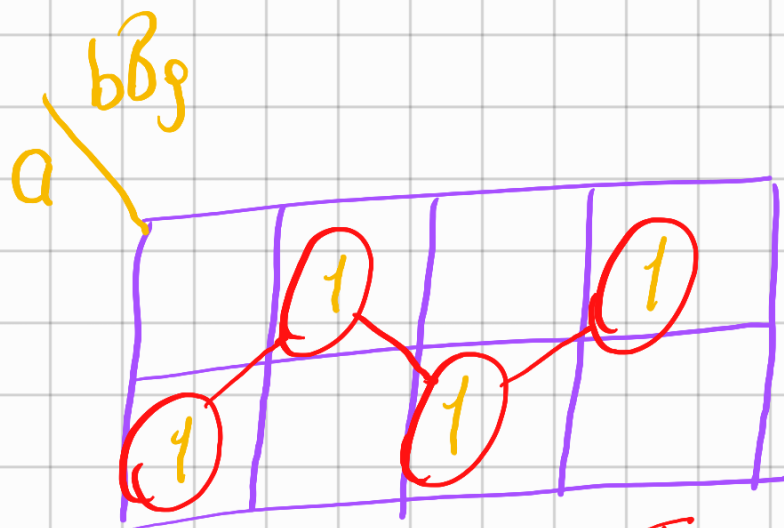
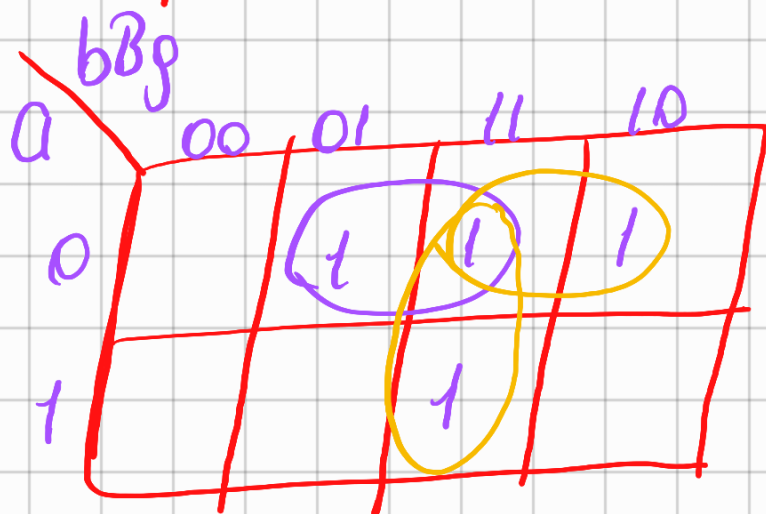
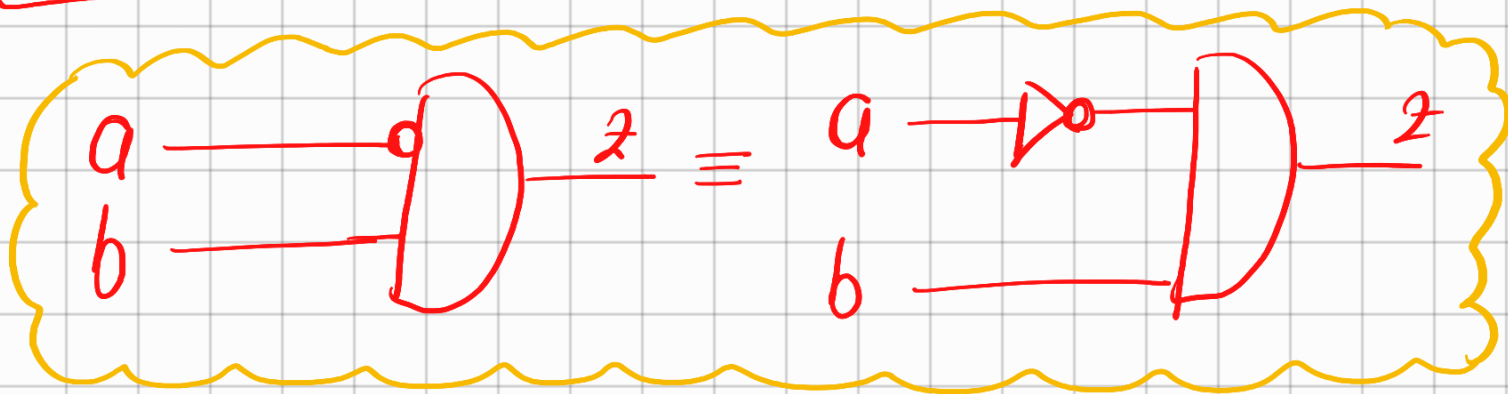
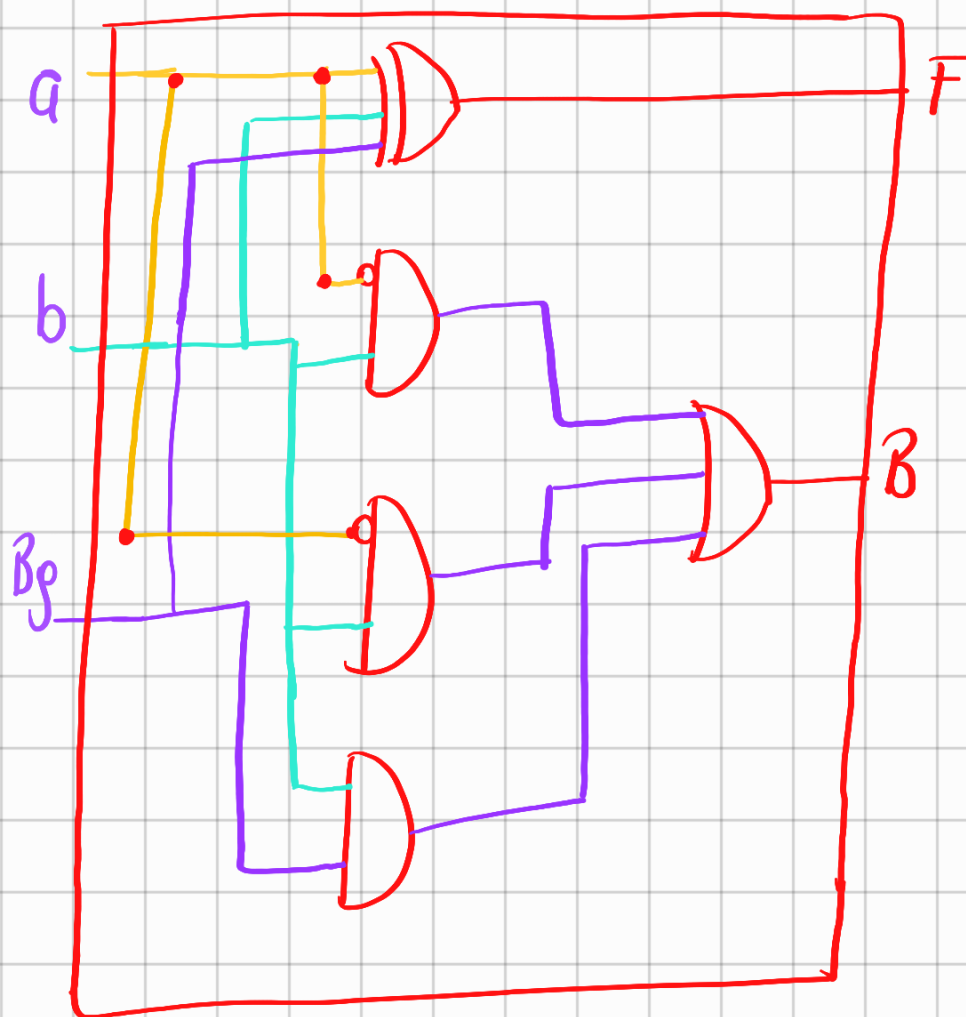


Fig 1

$$F = a \oplus b \oplus c$$



$$B = a'b + a'bc + bBc + Bc(a'+b)$$

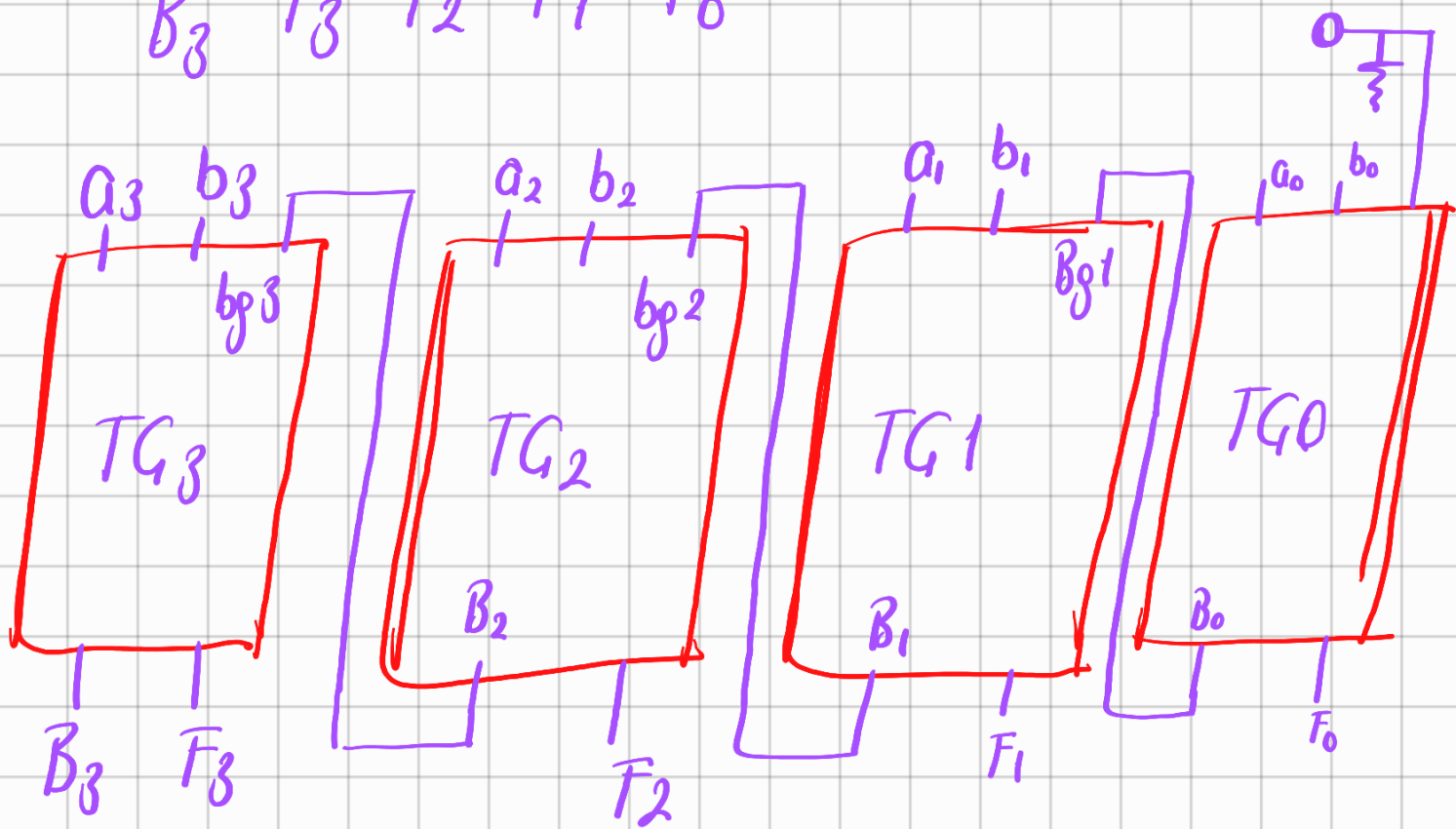


Örnek $\rightarrow$ 4 bitlik çıkarma

$a_3 \ a_2 \ a_1 \ a_0$

$b_3 \ b_2 \ b_1 \ b_0$

$B_3 \quad F_3 \quad F_2 \quad F_1 \quad F_0$



$$\begin{array}{r}
 00000 \\
 1111 \\
 \hline
 0001
 \end{array}$$

$$\begin{array}{r}
 11 \\
 22 \\
 \hline
 -89 \\
 \hline
 7
 \end{array}$$

3) toplama ve çıkarma devresi
(Adder/Subtractor)

a b c d
e f g h
=

ter
derresi)

a b c d
(e f g h) iki tüm
+

e' f' g' h'
+
→ ikiye tümleyen



